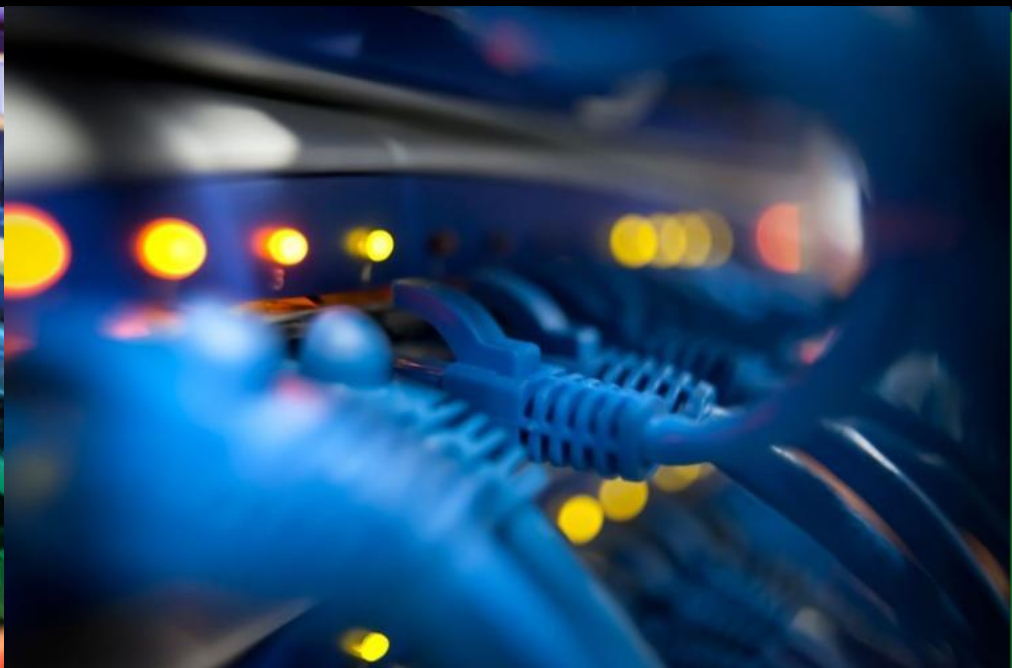




## AWS Lab: Configuring IAM, VPC, and EC2



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## Purpose

The purpose of this lab was to explore the more advanced capabilities of AWS, beyond basic EC2 and IAM functionalities. In this lab, we explored more of the cloud services provided by AWS such as AWS Elastic Block Storage (AWS EBS), Database Servers, and Load Balancers/Auto Scaling functionalities.

## Background

**Cloud computing**, as defined by the **International Organization for Standardization (ISO)** as "a paradigm for enabling network access to a scalable and elastic pool of shareable physical or virtual resources with self-service provisioning and administration on demand". It is commonly called "**the cloud**."

The cloud is a recent invention and has many differences and usages in comparison to on-premises networking. For example, **the cloud is on-demand**, meaning you could use computing capabilities at any time, with any amount, without requiring any human interaction with a service provider. Practically, this shifts the traditional on-premises networking that has immense capital/up-front expenditure **to a pay-as-you-go model with operational expenditures**.

**The cloud also maintains a global premise**, which allows you to deploy anywhere in the world within a few clicks. The immense number of data centers across the globe leased by the Cloud Platform (In this case, AWS), also ensures redundancy and high availability, whereas on premises would require duplicating hardware, which duplicates cost.

The cloud can be analogous to housing. Whereas buying one house can be expensive, and not fit your needs, renting a house of scalable size, which meets your needs, with a lease controlled majorly by you would grant scalability and availability. Despite this, on-premises still has benefits over cloud computing such as having ownership of hardware which leads to cost predictability at scale, no shared-tenancy concerns, and in general, a full autonomy over hardware and infrastructure.

**Amazon Web Services (AWS)** is a subsidiary of Amazon which provides said on-demand cloud computing on a metered, pay-as-you-go basis. One of the foundational services of AWS is Amazon EC2, which allows users to deploy a virtual cluster of computers with high availability. These virtual computers emulate most of the functionalities of an actual computer and are rented.

Amazon Identity and Access Management (IAM) is a web service that helps you control access to your AWS resources. Using IAM allows you to manage the permissions that control which AWS resources users can access. IAM allows you to authenticate and authorize people to use resources and provides said infrastructure.

IAM has different ways to enforce this, including IAM Policies, IAM Roles, IAM Groups, and IAM User. An IAM User represents an individual who uses AWS and has long-term credentials. By default, they have zero permissions. IAM Users may be collectively added into an IAM Group, who share the same permissions. Instead of attaching an IAM Policy to each user, you can attach it to a group. An AWS IAM Policy is a JSON document that explicitly defines what actions are allowed and not allowed on AWS. Like a user, an IAM role is similar, but is designed to be assumed temporarily for whoever needs it. It's commonly used to grant AWS services permission to interact with other AWS services.

Amazon Elastic Block Storage (EBS) is a service that allows you to create "Storage Volumes" and attach them to Amazon EC2 instances. Once attached, you can create a file system on top of these volumes, run a database, or use them just like you would use block storage. Amazon EBS volumes are placed in a specific Availability Zone, where they are automatically replicated to protect from the failure of a single component. All EBS volume types offer durable snapshot capabilities and are designed for high availability.

While traditionally, object storage is best for unstructured data and analytics, and file storage is best for shared access/enterprise file workloads, Block Storage is best for data bases, boot volumes, and low-latency workloads. And as the name suggests, while file storage stores data in accessible, hierarchical files/folders, block storage breaks the data into blocks that are stored independently, and it appears as a raw disk attached to a virtual machine.

Databases can be divided into two common categories, relational and non-relational. Relational databases organize information into tables with rows and columns and mostly use SQL to manage and access data. Oftentimes these tables will interact with other tables and are very ideal for transactional systems. Non-relational databases are more flexible, and store data in a variety of methods, including documents, key-value pairs, graphs, and more. Databases provide reliable, durable storage, which is needed because of their responsibilities, such as reading, writing, accessing, and managing data.

Scaling in Cloud Computing refers to changing compute capabilities based on need. Vertical Scaling would be to change the size or power of a single server by increasing its capabilities, such as more CPU, more RAM, more storage. As for horizontal scaling, it is to distribute a workload across an increased number of servers. Autoscaling in Cloud Computing is a feature that automatically adjusts computing resources based on the demand set. Instead of manually adding or removing compute resources, an auto scaler can dynamically scale or descale your infrastructure in real time based on the limits you set on it. For example, when a CPU threshold is met, new

EC2 instances can be launched automatically and then terminated when not needed.

Auto scaling is oftentimes combined with load balancing, which distributes workload evenly across all servers/instances to ensure that one server isn't being overworked and provides proper redundancy. Overall, monitoring, auto scaling, and load balancing work together to increase an architecture's fault tolerance, reliability, availability, and cost optimization.

## **Lab Summary:**

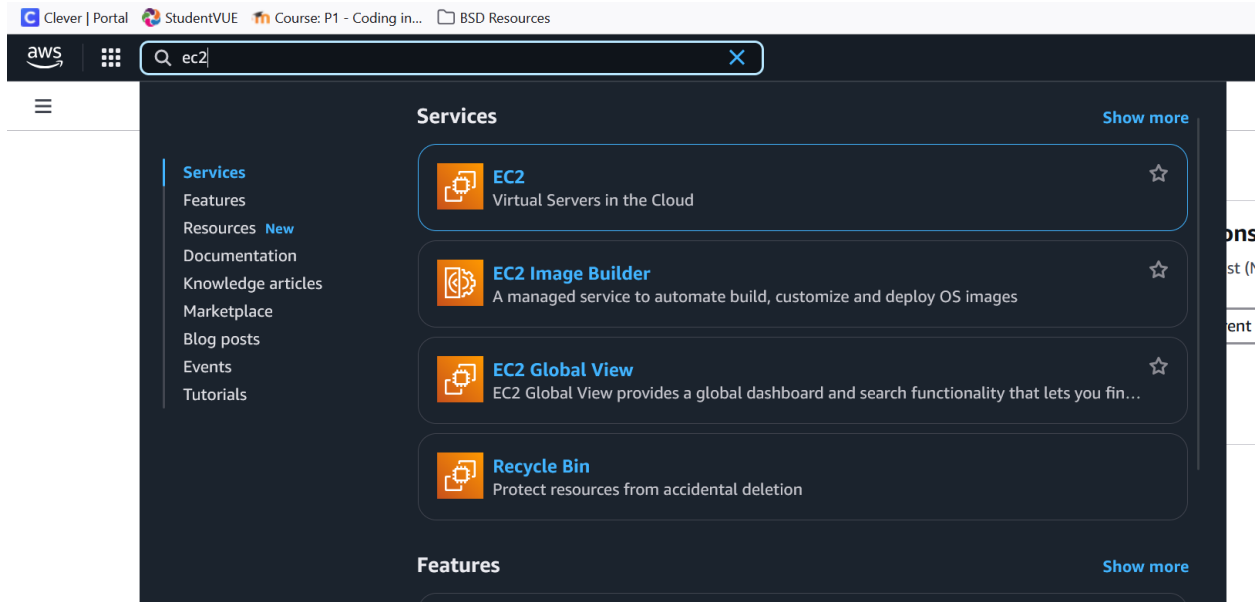
In this lab, we configured three sub-labs. The first explores AWS Elastic Block Storage, a unique storage system meant for EC2 Instances. The second involves setting up a data base server and changing contact information inside of it and increasing the indexes. The final includes setting up a CloudWatch alarm, such that when CPU utilization increases inside of AWS, EC2 Instances are temporarily created to support the increased workload.

## **Configurations:**

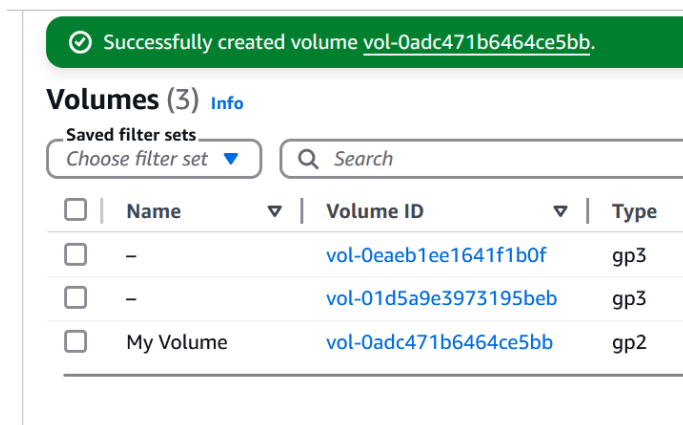
### **Lab 1: Working with AWS Elastic Block Storage (EBS)**

**Note:** To successfully complete this lab, you should be familiar with basic Amazon EC2 usage and with basic Linux server administration. You should feel comfortable using the Linux command-line tools.

1. In the "AWS Management Console", click "EC2"

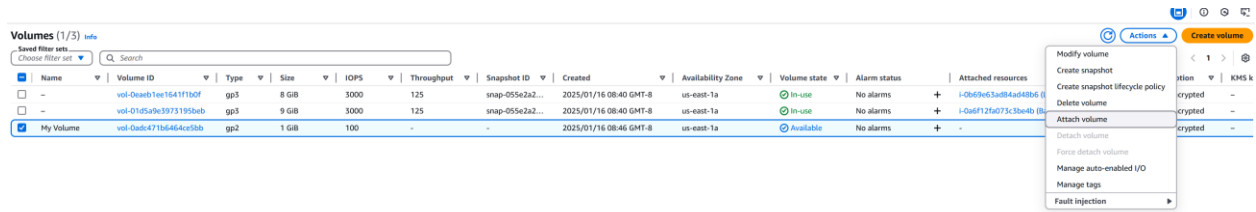


2. Choose “Volumes” then “Create Volume” and configure with the following:
  - Volume Type: General Purpose SSD (gp2)
  - Size (GiB): 1
  - Availability Zone: Select the same availability zone as your EC2 instance.
3. Choose “Add Tag” and configure with the following:
  - Key: Name
  - Value: My Volume
4. Choose “Create Volume”



5. Select “My Volume.”

6. In the “Actions” menu, choose “Attach volume.”



7. Choose the “Instance” field, then select the instance that appears (Lab).

8. For the Device Name, choose /dev/sdb/ from the dropdown.

**Note:** You will use this device identifier in a later task.

9. Choose “Attach Volume”

**Note:** The volume state is now In-use.

#### Attach volume [Info](#)

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

##### Basic details

###### Volume ID

vol-0adc471b6464ce5bb (My Volume)

###### Availability Zone

us-east-1a

###### Instance [Info](#)

i-0b69e63ad84ad48b6  
(Lab) (running)

Only instances in the same Availability Zone as the selected volume are displayed.

###### Device name [Info](#)

/dev/sdf

Recommended device names for Linux: /dev/xvda for root volume, /dev/sd[f-p] for data volumes.

Newer Linux kernels may rename your devices to /dev/xvdf through /dev/xvdp internally, even when the device name entered here (and shown in the details) is /dev/sdf through /dev/sdp.

[Cancel](#)

[Attach volume](#)

10. In the AWS Management Console, in the EC2 service, choose “Instances.”

11. Select the Lab instance.

12. Choose “Connect”

**Note:** A new browser tab will open with a terminal session connected to the instance.

Connect to your instance `i-0b69e63ad84ad48b6` (Lab) using any of these options

Cancel Connect

```

#_
~\_###_
~~\_#####\
~~\###|
~~\#/
~~V~' '->
~~~
~~~.~.~
~~~/_/~/
[m/'

[ec2-user@ip-10-1-11-4 ~]$
[ec2-user@ip-10-1-11-4 ~]$ df -h

```

| Filesystem   | Size | Used | Avail | Use% | Mounted on     |
|--------------|------|------|-------|------|----------------|
| devtmpfs     | 4.0M | 0    | 4.0M  | 0%   | /dev           |
| tmpfs        | 475M | 0    | 475M  | 0%   | /dev/shm       |
| tmpfs        | 190M | 452K | 190M  | 1%   | /run           |
| /dev/xvda1   | 8.0G | 1.6G | 6.4G  | 20%  | /              |
| tmpfs        | 475M | 0    | 475M  | 0%   | /tmp           |
| /dev/xvda128 | 10M  | 1.3M | 8.7M  | 13%  | /boot/efi      |
| tmpfs        | 95M  | 0    | 95M   | 0%   | /run/user/1000 |

```

[ec2-user@ip-10-1-11-4 ~]$

```

```
sudo su -l ec2-user
```



15. View the storage available on your instance:

```
df -h
```

**Note:** You should see output similar to:

| Filesystem | Size | Used | Avail | Use% | Mounted on     |
|------------|------|------|-------|------|----------------|
| devtmpfs   | 484M | 0    | 484M  | 0%   | /dev           |
| tmpfs      | 492M | 0    | 492M  | 0%   | /dev/shm       |
| tmpfs      | 492M | 460K | 491M  | 1%   | /run           |
| tmpfs      | 492M | 0    | 492M  | 0%   | /sys/fs/cgroup |
| /dev/xvda1 | 8.0G | 1.5G | 6.6G  | 19%  | /              |
| tmpfs      | 99M  | 0    | 99M   | 0%   | /run/user/0    |
| tmpfs      | 99M  | 0    | 99M   | 0%   | /run/user/1000 |

16. Create an ext3 file system on the new volume:

```
sudo mkfs -t ext3 /dev/sdb
```

17. Create a directory for mounting the new storage volume:

```
sudo mkdir /mnt/data-store
```

18. Mount the new volume:

```
sudo mount /dev/sdb /mnt/data-store
```

19. To configure the Linux instance to mount this volume whenever the instance is started, you will need to add a line to /etc/fstab.

```
echo "/dev/sdb    /mnt/data-store ext3 defaults,noatime 1 2" | sudo tee  
-a /etc/fstab
```



20. View the configuration file to see the setting on the last line:

```
cat /etc/fstab
```

21. View the available storage again:

```
df -h
```

**Note:** The output will now contain an additional line - **/dev/xvdb:**

22. On your mounted volume, create a file and add some text to it.

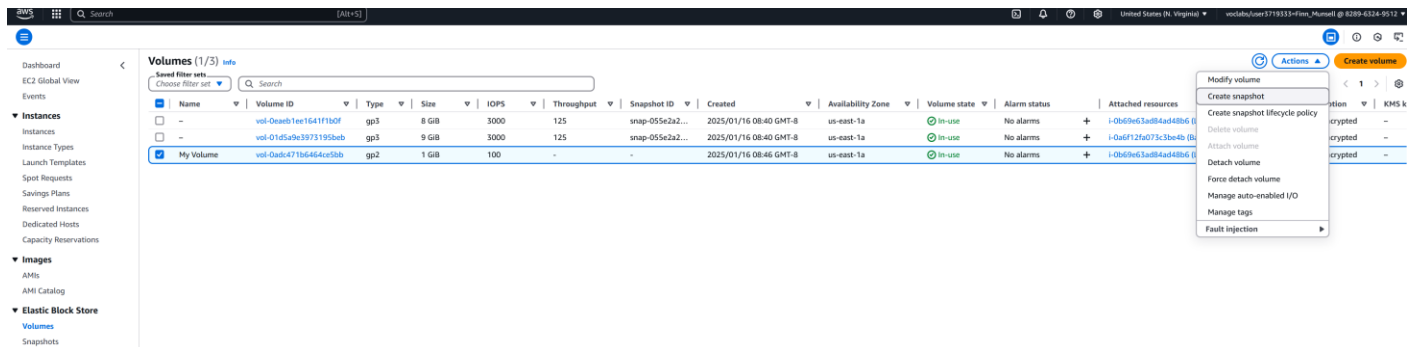
```
sudo sh -c "echo some text has been written > /mnt/data-store/file.txt"
```

23. Verify that the text has been written to your volume.

```
cat /mnt/data-store/file.txt
```

24. In the AWS Management Console, choose Volumes and select “My Volume”.

25. In the Actions menu, select “Create snapshot”.



26. Choose Add tag then configure:

- Key: Name
- Value: My Snapshot

27. Choose “Create snapshot”

28. In the left navigation pane, choose Snapshots.

**Note:** Your snapshot is displayed. The status will first have a state of Pending, which means that the snapshot is being created. It will then change to a state of Completed.

**Note:** Only used storage blocks are copied to snapshots, so empty blocks do not occupy any snapshot storage space.

29. In your terminal session, delete the file that you created on your volume.

```
sudo rm /mnt/data-store/file.txt
```

30. Verify that the file has been deleted.

```
ls /mnt/data-store/
```

**Note:** Your file has been deleted.

31. In the AWS Management Console, select My Snapshot.

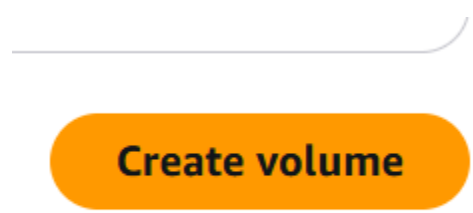
32. In the Actions menu, select Create volume from snapshot.

33. For Availability Zone Select the same availability zone that you used earlier.

34. Choose Add tag then configure:

- Key: Name
- Value: Restored Volume

35. Choose Create volume



Create volume

**Note:** *When restoring a snapshot to a new volume, you can also modify the configuration, such as changing the volume type, size or Availability Zone.*

36. In the left navigation pane, choose Volumes.

37. Select Restored Volume.

38. In the Actions menu, select Attach volume.

39. Choose the Instance field, then select the (Lab) instance that appears.

40. For the Device Name, choose /dev/sdc/ (or any next available identifier) from the dropdown.

41. Choose Attach volume

**Note:** *The volume state is now in use.*

42. Create a directory for mounting the new storage volume:

```
sudo mkdir /mnt/data-store2
```

43. Mount the new volume:

```
sudo mount /dev/sdc /mnt/data-store2
```

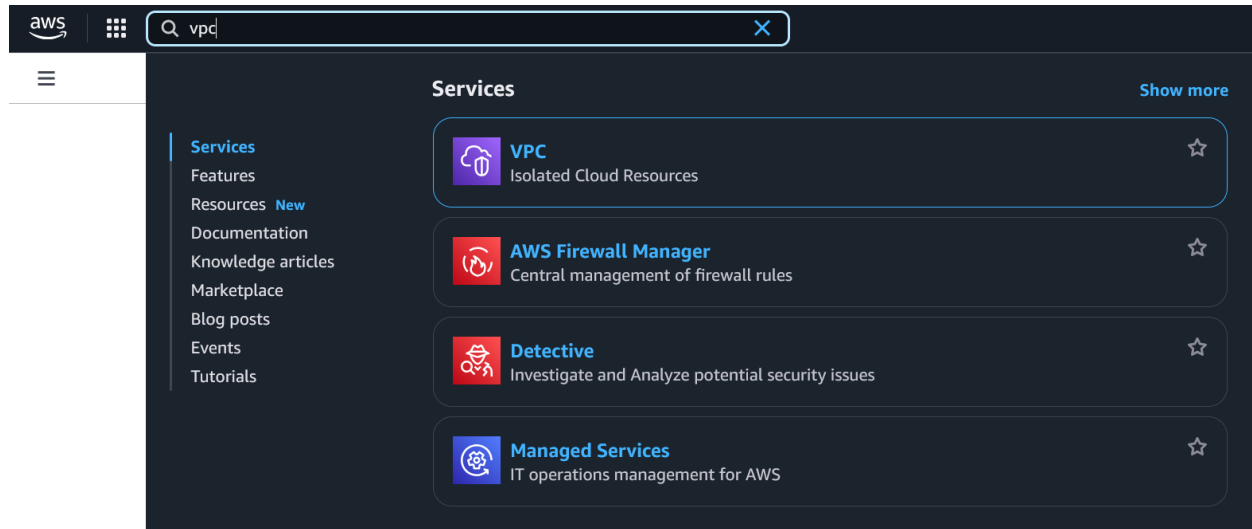
44. Verify that volume you mounted has the file that you created earlier.

```
ls /mnt/data-store2/
```

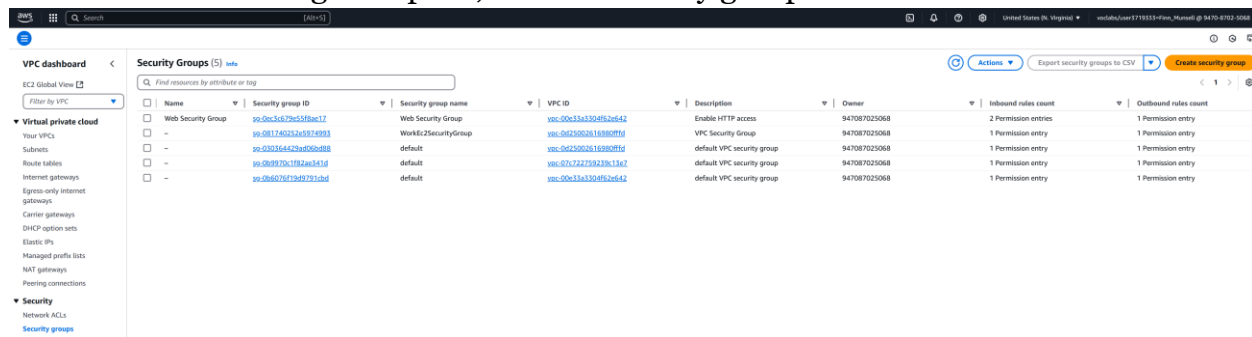
**Note:** *You should see file.txt.*

## **Lab 2: Build a Database Server**

1. In the AWS Management Console, in the search box next to Services, search for and select VPC.



2. In the left navigation pane, choose Security groups.



3. Choose Create security group and then configure:
  - Security group name: DB Security Group
  - Description: Permit access from Web Security Group
  - VPC: Lab VPC
  - Tip: Choose the X next to VPC that is already selected, then choose Lab VPC from the menu.

4. In the Inbound rules pane, choose Add rule

Note: The security group currently has no rules. You will add a rule to permit access from the Web Security Group.

5. Configure the following settings:

- Type: MySQL/Aurora (3306)
- Source: Place your cursor in the field to the right of Custom, type sg, and then select Web Security Group.

**Note:** This configures the Database security group to permit inbound traffic on port 3306 from any EC2 instance that is associated with the Web Security Group.

6. Choose Create security group

**Note:** You will use this security group when launching an Amazon RDS database in this lab.

The screenshot shows the AWS Management Console interface for a security group. At the top, a green notification bar states: "Security group (sg-04ddc17d791e548af | DB Security Group) was created successfully". Below this, the title "sg-04ddc17d791e548af - DB Security Group" is displayed. The "Details" section contains the following information:

| Details                                         |                                                  |                                                             |                                        |
|-------------------------------------------------|--------------------------------------------------|-------------------------------------------------------------|----------------------------------------|
| <b>Security group name</b><br>DB Security Group | <b>Security group ID</b><br>sg-04ddc17d791e548af | <b>Description</b><br>Permit access from Web Security Group | <b>VPC ID</b><br>vpc-00e33a3304f62e642 |
| <b>Owner</b><br>947087025068                    | <b>Inbound rules count</b><br>1 Permission entry | <b>Outbound rules count</b><br>1 Permission entry           |                                        |

Below the details, there are tabs for "Inbound rules", "Outbound rules", "Sharing - new", "VPC associations - new", and "Tags". The "Inbound rules" tab is selected, showing a table with one rule:

| Name | Security group rule ID | IP version | Type         | Protocol | Port range | Source                  |
|------|------------------------|------------|--------------|----------|------------|-------------------------|
| -    | sgr-016a23b44a7579b9b  | -          | MySQL/Aurora | TCP      | 3306       | sg-0ec3c679e55f8ae17... |

7. In the AWS Management Console, in the search box next to Services, search for and select RDS.

The screenshot shows the AWS Management Console search results for "RDS". The search bar at the top contains "RDS". The left navigation pane shows "VPC" selected. The main content area displays a list of services:

- RDS** (Managed Relational Database Service)
- Database Migration Service** (Managed Database Migration Service)
- Kinesis** (Work with Real-Time Streaming Data)
- Amazon OpenSearch Service** (Run open-source OpenSearch or Elasticsearch using Managed Clusters or Serverless de...)

8. In the left navigation pane, choose Subnet groups.

9. Choose Create DB Subnet Group then configure:

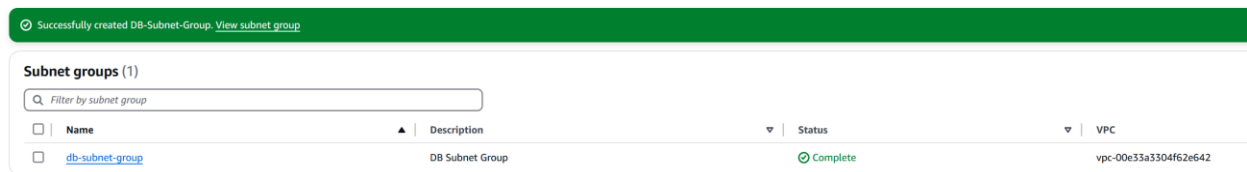
- **Name:** DB-Subnet-Group
- **Description:** DB Subnet Group
- **VPC:** *Lab VPC*

10. Scroll down to the Add subnets section.

11. Expand the list of values under Availability Zones and select the first two zones: us-east-1a and us-east-1b.

12. Expand the list of values under Subnets and select the subnets associated with the CIDR ranges 10.0.1.0/24 and 10.0.3.0/24.

13. Choose Create



14. In the left navigation pane, choose Databases.

15. Choose Create database

16. Select MySQL under Engine Options.

17. Under Templates, choose Dev/Test.

18. Under Availability and durability, choose Multi-AZ DB instance.

19. Under Settings, configure:

- **DB instance identifier:** lab-db
- **Master username:** main

- **Master password:** lab-password
- **Confirm password:** lab-password

20. Under **DB instance class**, configure:

- Select **Burstable classes (includes t classes)**.
- Select *db.t3.micro*

21. Under **Storage**, configure:

- Storage type: General Purpose (SSD)
- Allocated storage: 20

22. Under **Connectivity**, configure:

- Virtual Private Cloud (VPC): Lab VPC

23. Under Existing VPC security groups, from the dropdown list:

- Choose DB Security Group.
- Deselect default.

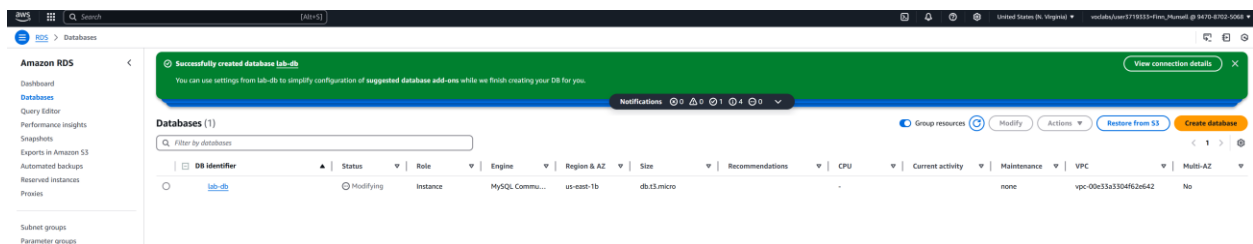
24. Under Monitoring expand Additional configuration.

- Uncheck Enable Enhanced monitoring.

25. Under Additional configuration, configure:

- Initial database name: lab
- Uncheck Enable automatic backups.
- Uncheck Enable encryption

26. Choose Create database



27. Choose lab-db (choose the link itself).



28. Wait until Info changes to Modifying or Available.

29. Scroll down to the Connectivity & security section and copy the Endpoint field.

- It will look similar to: lab-db.xxxx.us-east-1.rds.amazonaws.com.

30. Paste the Endpoint value into a text editor. You will use it later in the lab.

31. To discover the WebServer IP address, choose on the AWS Details drop down menu above these instructions. Copy the IP address value.

00:30

▶ Start Lab

■ End Lab

ⓘ AWS Details

ⓘ Details

✕

Submit

Submission Report

Grades

Close

Cloud Access

AWS CLI:

Show

Cloud Labs

Remaining session time: 00:29:04(30 minutes)

Session started at: 2025-01-17T07:56:16-0800

Session to end at: 2025-01-17T09:30:07-0800

Accumulated lab time: 01:04:00 (64 minutes)

(1) ips -- public:54.242.201.42, private:10.0.0.80 (2) ips -- public:52.90.168.167, private:10.0.2.49

SSH key

Show

Download PEM

Download PPK

AWS SSO

Download URL

|             |                                          |
|-------------|------------------------------------------|
| SecretKey   | u/AUud1iD/PfMa23ZMGV6j3PD1N6MxGClp7w8/lw |
| WebServer   | 52.90.168.167                            |
| BastionHost | 54.242.201.42                            |
| Region      | us-east-1                                |
| AccessKey   | AKIA5ZAWJEOWHBIAXIFU                     |

32. Open a new web browser tab, paste the WebServer IP address, and press Enter.

33. Choose the RDS link at the top of the page.

34. Configure the following settings:

- Endpoint: Paste the Endpoint you copied to a text editor earlier
- Database: lab
- Username: main
- Password: lab-password
- Choose Submit

**Note:** A message will appear explaining that the application is running a command to copy information to the database. After a few seconds, the application will display an Address Book. The Address Book application is using the RDS database to store information.

 Load Test RDS

Endpoint

lab-db.cviqrvtzdyj.us-east-1.rds.amazonaws.com

Database

lab

Username

main

Password

••••••••••

Submit

35. Test the web application by adding, editing, and removing contacts.

**Note:** The data is being persisted to the database and is automatically replicating to the second Availability Zone.

Address Book

Data Updated!

| Last name | First name | Phone        | Email                                                                              | Admin                                       |
|-----------|------------|--------------|------------------------------------------------------------------------------------|---------------------------------------------|
|           |            |              |                                                                                    | <a href="#">Add Contact</a>                 |
| Johnson   | Roberto    | 123-456-7890 | <a href="mailto:roberto@someaddress.com">roberto@someaddress.com</a>               | <a href="#">Edit</a> <a href="#">Remove</a> |
| Keanu     | Reeves     | 010-110-1101 | <a href="mailto:duuuuuude@someotheraddress.org">duuuuuude@someotheraddress.org</a> | <a href="#">Edit</a> <a href="#">Remove</a> |

## Lab 3: Scale & Load Balance Architecture

1. In the AWS Management Console, in the search box next to Services, search for and select EC2
2. In the left navigation pane, choose Instances.
3. Wait until the Status Checks for Web Server 1 displays 2/2 checks passed. If necessary, choose refresh to update the status.
4. Select Web Server 1.
5. In the Actions menu, choose Image and templates > Create image, then configure:
  - Image name: WebServerAMI
  - Image description: Lab AMI for Web Server
6. Choose Create image

**Note:** *A confirmation banner displays the AMI ID for your new AMI.*

You will use this AMI when launching the Auto Scaling group later in the lab.

7. In the left navigation pane, choose Target Groups.
  - Choose Create target group
  - Choose a target type: Instances
  - Target group name, enter: LabGroup
  - Select Lab VPC from the VPC drop-down menu.
8. Choose Next. The Register targets screen appears.

**Note:** *Targets are the individual instances that will respond to requests from the Load Balancer. You do not have any web application instances yet, so you can skip this step.*

9. Review the settings and choose Create target group
10. In the left navigation pane, choose **Load Balancers**.

11. At the top of the screen, choose Create load balancer.

12. Under Application Load Balancer, choose Create

13. Under Load balancer name, enter: LabELB

**Basic configuration**

**Load balancer name**  
Name must be unique within your AWS account and can't be changed after the load balancer is created.

LabELB

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

**Scheme** [Info](#)  
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the **IPv4** and **Dualstack** IP address types.

**Load balancer IP address type** [Info](#)  
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address types. Public IPv4 address:

☒ **IPv4**  
Includes only IPv4 addresses.

☐ **Dualstack**  
Includes IPv4 and IPv6 addresses.

☐ **Dualstack without public IPv4**  
Includes a public IPv6 address, and private IPv4 and IPv6 addresses. Compatible with **internet-facing** load balancers only.

14. Scroll down to the Network mapping section, then:

- For VPC, choose Lab VPC
- You will now specify which subnets the Load Balancer should use. The load balancer will be internet-facing, so you will select both Public Subnets.
- Choose the first displayed Availability Zone, then select Public Subnet 1 from the Subnet drop down menu that displays beneath it.
- Choose the second displayed Availability Zone, then select Public Subnet 2 from the Subnet drop down menu that displays beneath it.
- You should now have two subnets selected: Public Subnet 1 and Public Subnet 2.

### Network mapping [Info](#)

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address settings.

#### VPC [Info](#)

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets must be hosted unless routing to Lambda or on-premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-0c24aa91f54369a14 (Lab VPC)  
10.0.0.0/16



[Create VPC](#)

#### IP pools [Info](#)

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or view [Pools](#) in the [Amazon VPC IP Address Manager console](#).

☐ Use IPAM pool for public IPv4 addresses

The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses will be assigned by AWS.

#### Availability Zones and subnets [Info](#)

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone and will automatically scale in response to traffic. The load balancer routes traffic to targets in the selected Availability Zones only.

☒ **us-east-1a (use1-az2)**

##### Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-02417564a11af8bed (Public Subnet 1)  
10.0.0.0/24

☒ **us-east-1b (use1-az4)**

##### Subnet

Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are required for your load balancer to scale efficiently.

subnet-0a4274c58bce221ea (Public Subnet 2)  
10.0.2.0/24

### 15. In the Security groups section:

- Choose the Security groups drop down menu and select Web Security Group
- Below the drop-down menu, choose the X next to the default security group to remove it.
- The Web Security Group security group should now be the only one that appears.

## Security groups [Info](#)

A security group is a set of firewall rules that control the traffic to your load balancer. Select an existing security group, or you can [create a new security group](#).

### Security groups

Select up to 5 security groups

Web Security Group (sg-0ccfed20b393abbc) X

## Listeners and routing [Info](#)

A listener is a process that checks for connection requests using the port and protocol you configure. The rules that you define for a listener determine how the load balancer routes requests to its registered targets.

▼ Listener HTTP:80

Remove

Protocol

HTTP

Port

80

1-65535

Default action [Info](#)

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action

☒ Forward to target groups

☐ Redirect to URL

☐ Return fixed response

Forward to target group [Info](#)

Choose a target group and specify routing weight or [create target group](#).

Target group

LabGroup

Target type: Instance, IPv4 | Target stickiness: Off

HTTP

Weight

1

0-999

Percent

100%

+ Add target group

You can add up to 4 more target groups.

Target group stickiness [Info](#)

Enables the load balancer to bind a user's session to a specific target group. To use stickiness the client must support cookies. If you want to bind a user's session to a specific target, turn on the Target Group attribute Stickiness.

☐ Turn on target group stickiness

Listener tags - optional

Consider adding tags to your listener. Tags enable you to categorize your AWS resources so you can more easily manage them.

Add listener tag

You can add up to 50 more tags.

16. For the Listener HTTP:80 row, set the Default action to forward to LabGroup.

17. Scroll to the bottom and choose Create load balancer

- Choose View load balancer
- The load balancer will show a state of provisioning. There is no need to wait until it is ready. Please continue with the next task.

18. In the left navigation pane, choose Launch Templates.

19. Choose Create launch template

20. Configure the launch template settings and create it:

- Launch template name: LabConfig
- Under Auto Scaling guidance, select Provide guidance to help me set up a template that I can use with EC2 Auto Scaling
- In the Application and OS Images (Amazon Machine Image) area, choose My AMIs.
- Amazon Machine Image (AMI): choose Web Server AMI
- Instance type: choose t2.micro
- Key pair name: choose vockey
- Firewall (security groups): choose Select existing security group
- Security groups: choose Web Security Group
- Scroll down to the Advanced details area and expand it.
- Scroll down to the Detailed CloudWatch monitoring setting. Select Enable

**Note:** *This will allow Auto Scaling to react quickly to changing utilization.*

- Choose Create launch template

## Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

**Launch template name and description**  
Launch template name - *required*  
  
Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '~', '@'.

**Template version description**  
  
Max 255 chars

**Auto Scaling guidance** | [Info](#)  
Select this if you intend to use this template with EC2 Auto Scaling  
☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

[▶ Template tags](#)  
[▶ Source template](#)

## Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

**▼ Application and OS Images (Amazon Machine Image) - required** [Info](#)  
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Recents

**My AMIs**

Quick Start

☒ Owned by me

☐ Shared with me

[Browse more AMIs](#)  
Including AMIs from AWS, Marketplace and the Community

**Amazon Machine Image (AMI)**  

WebServerAMI  
ami-0eaf9250bc25b0804  
2026-03-04T18:20:31.000Z    Virtualization: hvm    ENA enabled: true  
Root device type: ebs    Boot mode: uefi-preferred

**Description**  
Lab AMI for Web Server

| Architecture | AMI ID                |
|--------------|-----------------------|
| x86_64       | ami-0eaf9250bc25b0804 |

21. In the Success dialog, choose the LabConfig launch template.

22. From the Actions menu, choose Create Auto Scaling group

23. Configure the details in Step 1 (Choose launch template or configuration):

- Auto Scaling group name: Lab Auto Scaling Group
- Launch template: confirm that the LabConfig template you just created is selected.



- Choose Next

24. Configure the details in Step 2 (Choose instance launch options):

- VPC: choose Lab VPC
- Availability Zones and subnets: Choose Private Subnet 1 and then choose Private Subnet 2.
- Choose Next

25. Configure the details in Step 3 (Configure advanced options):

- Choose Attach to an existing load balancer
- Existing load balancer target groups: select LabGroup.
- In the Additional settings pane:
- Select Enable group metrics collection within CloudWatch
- This will capture metrics at 1-minute intervals, which allows Auto Scaling to react quickly to changing usage patterns.
- Choose Next

26. Configure the details in Step 4 (Configure group size and scaling policies - optional):

- Under Group size, configure:
  - Desired capacity: 2
  - Minimum capacity: 2
  - Maximum capacity: 6

**Note:** *This will allow Auto Scaling to automatically add/remove instances, always keeping between 2 and 6 instances running.*

- Under Scaling policies, choose Target tracking scaling policy and configure:
  - Scaling policy name: LabScalingPolicy
  - Metric type: Average CPU Utilization
  - Target value: 60

**Note:** *This tells Auto Scaling to maintain an average CPU utilization across all instances at 60%. Auto Scaling will automatically add or remove capacity as required to keep the metric at, or close to, the specified target value. It adjusts to fluctuations in the metric due to a fluctuating load pattern.*

- Choose Next

27. Configure the details in Step 5 (Add notifications - optional):

- Choose Next

**Note:** *Auto Scaling can send a notification when a scaling event takes place. You will use the default settings.*

28. Configure the details in Step 6 (Add tags - optional):

**Note:** *Tags applied to the Auto Scaling group will be automatically propagated to the instances that are launched.*

- Choose Add tag and configure the following:
  - Key: Name
  - Value: Lab Instance
- Choose Next

29. Configure the details in Step 6 (Review):

- Review the details of your Auto Scaling group
- Choose Create Auto Scaling group

**Note:** *Your Auto Scaling group will initially show an instance count of zero, but new instances will be launched to reach the Desired count of 2 instances.*

### Additional settings

|                                                                 |                              |                                            |
|-----------------------------------------------------------------|------------------------------|--------------------------------------------|
| <b>Instance scale-in protection</b><br>Disabled                 | <b>Monitoring</b><br>Enabled | <b>Default instance warmup</b><br>Disabled |
| <b>Auto Scaling group deletion protection</b><br>None (default) |                              |                                            |

### Capacity Reservation preference

|                              |                                      |                             |
|------------------------------|--------------------------------------|-----------------------------|
| <b>Preference</b><br>Default | <b>Capacity Reservation IDs</b><br>- | <b>Resource Groups</b><br>- |
|------------------------------|--------------------------------------|-----------------------------|

#### Step 5: Add notifications

##### Notifications

No notifications

#### Step 6: Add tags

##### Tags (1)

| Key  | Value        | Tag new instances |
|------|--------------|-------------------|
| Name | Lab Instance | Yes               |

[Preview code](#)
[Cancel](#)
[Previous](#)
[Create Auto Scaling group](#)

30. In the left navigation pane, choose Instances.

**Note:** You should see two new instances named *Lab Instance*. These were launched by *Auto Scaling*.

If the instances or names are not displayed, wait 30 seconds and choose refresh in the top-right.

Next, you will confirm that the new instances have passed their Health Check.

31. In the left navigation pane, choose Target Groups.

32. Select LabGroup

33. Choose the Targets tab.

34. Wait until the Status of both instances' transitions to healthy.

35. In the left navigation pane, choose Load Balancers.

36. Select the LabELB load balancer.

37. In the Details pane, copy the DNS name of the load balancer, making sure to omit "(A Record)".

38. Open a new web browser tab, paste the DNS Name you just copied, and press Enter.

39. Return to the AWS Management Console, but do not close the application tab — you will return to it soon.

40. In the search box next to Services, search for and select CloudWatch.

41. In the left navigation pane, choose All alarms.

**Note:** *Two alarms will be displayed. These were created automatically by the Auto Scaling group. They will automatically keep the average CPU load close to 60% while also staying within the limitation of having two to six instances.*

42. Choose the OK alarm, which has AlarmHigh in its name.

43. Return to the browser tab with the web application.

44. Choose Load Test beside the AWS logo.

**Note:** *This will cause the application to generate high loads. The browser page will automatically refresh so that all instances in the Auto Scaling group will generate load. Do not close this tab.*

45. Return to the browser tab with the CloudWatch console.

46. Wait until the AlarmHigh alarm enters the In alarm state.

**Note:** *You can now view the additional instance(s) that were launched.*

47. In the search box next to Services, search for and select EC2.

48. In the left navigation pane, choose Instances.

**Note:** *More than two instances labeled Lab Instance should now be running. The new instance(s) were created by Auto Scaling in response to the CloudWatch alarm.*

49. Select Web Server 1 (and ensure it is the only instance selected).

50. In the Instance state menu, choose Instance State > Terminate Instance.

51. Choose Terminate

## **Problems:**

There were no major problems in this lab.

## **Conclusion:**

This lab was great at teaching more advanced concepts in AWS, and it was a terrific way to learn how these concepts come together in the actual AWS interface. I am now confident in my skills to apply all these concepts in a real-life practical and theoretical way.